

Artificial Intelligence — Beginner Learning Guide

EduNova AI Course Material

This document introduces the basic ideas of Artificial Intelligence (AI), Machine Learning (ML), neural networks, and responsible AI in simple language.

1. What is Artificial Intelligence?

Artificial Intelligence is a field of computer science that creates systems that can perform tasks that normally require human intelligence. These tasks include understanding language, recognizing images, finding patterns, making predictions, and assisting with decisions.

Example: A voice assistant can understand a spoken question and provide a response.

2. What is Machine Learning?

Machine Learning is a part of AI in which computers learn patterns from data instead of being explicitly programmed with every rule.

Example: A spam filter learns from examples of spam and normal emails.

3. Types of Machine Learning

Supervised Learning: The model learns from labelled examples.

Unsupervised Learning: The model finds patterns or groups in data without labels.

Reinforcement Learning: An agent learns by receiving rewards or penalties for actions.

4. Neural Networks

A neural network is a machine-learning model made of connected layers of artificial neurons. Information moves through the network and the model adjusts its internal parameters during training.

Common structure: Input Layer → Hidden Layers → Output Layer

5. Generative AI

Generative AI creates new content such as text, images, audio, or code based on patterns learned from training data.

Example: A language model can generate an explanation of a programming concept.

6. Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with a language model. First, relevant information is retrieved from a knowledge base. Then the language model uses that information to generate an answer.

Basic flow: Document → Chunking → Retrieval → Relevant Context → AI Answer

7. Source Grounding

Source grounding means connecting an AI answer to the material used to produce the answer. A useful learning system can show the document name and page, slide, or timestamp where the information came from.

8. Adaptive Learning

Adaptive learning changes questions and learning recommendations based on a learner's previous answers and topic mastery. For example, a learner who repeatedly struggles with one concept can receive additional practice on it.

9. Responsible AI

AI systems should be designed with attention to accuracy, privacy, fairness, security, transparency, and appropriate human oversight.

10. Quick Review Questions

1. What does AI stand for?
2. How is Machine Learning related to AI?
3. What is supervised learning?
4. What is RAG?
5. Why are source citations useful in an AI learning system?